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Smallholder maize area and yield mapping at national scales with Google Earth Engine

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Highlights

- National maps of crop and maize areas are generated at 10 m resolution in Africa.
- Maize yields are mapped accurately without any ground calibration.
- Independent soil data can explain over 70% of variability in yield estimates.
- Soil nitrogen and organic carbon exhibit the strongest association with yields.
- [Google Earth Engine](#) and Sentinel data offer new opportunities for mapping agriculture.

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Abstract

Accurate measurements of maize yields at field or subfield scales are useful for guiding agronomic practices and investments and policies for improving food security. Data on [smallholder](#) maize systems are currently sparse, but [satellite remote sensing](#) offers promise for accelerating learning about these systems. Here we document the use of [Google Earth Engine](#) (GEE) to build “wall-to-wall” 10 m resolution maps of (i) cropland presence, (ii) maize presence, and (iii) maize yields for the main 2017 maize season in Kenya and Tanzania. Mapping these outcomes at this scale is extremely challenging because of very

heterogeneous landscapes, lack of cloud-free satellite imagery, and the low quantity of quality ground-based data in these regions.

First, we computed seasonal median composites of Sentinel-1 radar backscatter and Sentinel-2 optical reflectance measures for each pixel in the region, and used them to build both crop/non-crop and maize/non-maize Random Forest (RF) classifiers. Several thousand crop/non-crop labels were collected through an in-house GEE labeler, and thousands of crop type labels from the 2015–2017 growing seasons were obtained from various sources. Results show that the crop/non-crop classifier successfully identified cropland with over 85% out-of-sample accuracy in both countries, with Sentinel-1 being particularly useful for prediction. Among the cropped pixels, the maize/non-maize classifier had an accuracy of 79% in Tanzania and 63% in Kenya.

To map maize yields, we build on past work using a scalable crop yield mapper (SCYM) that utilizes simulations from a crop model to train a regression that predicts yields from observations. Here we advance past approaches by (i) grouping simulations by Global Agro-Environmental Stratification (GAES) zones across the two countries, in order to account for landscape heterogeneity, (ii) utilizing gridded datasets on soil and sowing and harvest dates to setup model simulations in a scalable way; and (iii) utilizing all available satellite observations during the growing season in a parsimonious way by using harmonic regression fits implemented in GEE. SCYM estimates were able to capture about 50% of the variation in the yields at the district level in Western Kenya as measured by objective ground-based crop cuts.

Finally, we illustrated the utility of our yield maps with two case studies. First, we document the magnitude and interannual variability of spatial heterogeneity of yields in each district, and how it varies for different parts of the region. Second, we combine our estimates with recently released soil databases in the region to investigate the most important soil constraints in the region. Soil factors explain a high fraction (72%) of variation in predicted yields, with the predominant factor being soil nitrogen levels. Overall, this study illustrates the power of combining Sentinel-1 and Sentinel-2 imagery, the GEE platform, and advanced classification and yield mapping algorithms to advance understanding of smallholder agricultural systems.

Keywords

Sentinel-1; Sentinel-2; Data fusion; Cropland classification; Yield mapping

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Introduction

Smallholder farms are the most common form of agriculture in the world, particularly in food insecure regions of South Asia and Sub-Saharan Africa

(Lowder et al., 2016; Samberg et al., 2016). For example, recent analysis of sub-national census data suggests that over 90% of food calories in South Asia and 50% in Sub-Saharan Africa originate from farms smaller than 5 ha in size (Samberg et al., 2016). Yet much remains unknown about these smallholder systems, including basic information on how efficiently they use key inputs such as land and labor, and how these productivities compare to those on larger farms or the profitability of off-farm activities (Carletto et al., 2015; Mccullough, 2017). The need for improved data on smallholder systems has motivated renewed investment in rigorous ground surveys such as the World Bank's Living Standards Measurement Study – Integrated Surveys on Agriculture (Carletto et al., 2015), as well as interest in the potential use of satellite remote sensing to provide low-cost measures of some key aspects of smallholder farms (Burke and Lobell, 2017; Chivasa et al., 2017).

New satellite sensors with spatial resolutions of 10 m or finer have opened up new possibilities for monitoring smallholder fields. For example, a 1 ha field consists of only ~10 pixels in a Landsat image with 30 m resolution, making it extremely difficult to delineate fields. In contrast, that same field would contain ~100 pixels in an image from Sentinel-2 with 10 m resolution, or ~1000 pixels in an image from a Planetscope sensor with roughly 3 m resolution. Given the 5-day revisit of Sentinel-2 since early 2017 and the roughly daily re-visit of Planetscope since late 2017, it is now possible to observe smallholder fields at a much higher frequency than ever before.

However, several aspects of smallholder farming, beyond the small sizes of fields, are challenging in the context of remote sensing. First and foremost is the ubiquitous cloud cover in many smallholder regions, which reside predominantly in the tropics and are actively growing crops during the rainy season (which is also the cloudy season). Second is the tremendous heterogeneity of most smallholder systems, with families typically growing multiple crops, often interspersed with each other and often including tree crops such as avocado or mangoes. In addition to this within-farm heterogeneity, there are often large differences between the crops grown by smallholders in one region and by farmers in a neighboring district or state.

Despite these obstacles, recent work has shown that images from Sentinel-2, Planet, and other fine-resolution sensors can be used to provide yield estimates on individual fields for a range of crops including maize, wheat, and millet (Jain et al., 2016; Burke and Lobell, 2017; Jin et al., 2017a; Lambert et al., 2018). These examples were typically from small sub-regions of ~100 km² or less, often chosen based on the availability of reliable ground-based data on yields, such as from farmer surveys or on-farm crop cuts, as well as the existence of cloud-free optical data. Moreover, because these studies typically compared yield estimates for specific fields with ground-based data, they did not have to address one of the main challenges of heterogeneous landscapes, which is identifying where specific crops or mixture of crops are growing. An exception is Jin et al. (2017a), who performed a classification across 40,000 km² in the Western Kenya region using Sentinel-2 data in order to map maize areas and yields, which were then

aggregated to the district scale for comparison with averages of on-farm crop cuts.

To make remote sensing a more reliable tool for measuring smallholder yields, a remaining challenge is to develop methods that are truly scalable to the national and global level. That is, they will work in any smallholder region in any year, with as little dependence as possible on obtaining cloud-free conditions or having access to reliable ground measurements for calibration. Here we describe the development of methods towards this end, building on prior work in Eastern Africa (Burke and Lobell, 2017; Jin et al., 2017a). The resulting approach is illustrated for the cases of Kenya and Tanzania, for which country-wide maps of maize yields at 10 m resolution are generated for 2017. We also illustrate two examples of how these types of datasets can help advance understanding of yield constraints in smallholder systems.

The amount of data required for this task is substantial, with the countries of Kenya (580,367 km²) and Tanzania (947,303 km²) together comprising over 1.5 million km², and with this study using both Sentinel-1 and Sentinel-2 data at 10 m resolution. Accounting for the 6-day revisit period of Sentinel-1 and 5-day revisit of Sentinel-2 at the equator, this translates to roughly 200 billion pixels of observations per month over the study region, or roughly 1 trillion over the course of a growing season. To access and process this volume of data, we rely on the unique capabilities of the Google Earth Engine (GEE) cloud-based platform (Gorelick et al., 2017).

The following section describes the data inputs used for this study, as well as a customized model for masking clouds and cloud shadows in Sentinel-2 for this region, given the relatively poor performance of the default cloud mask in this region. Section 3 describes the approach to mapping the presence of field crops in general, and maize in particular, which are important precursors to mapping maize yields in the region. Section 4 then presents an updated approach to mapping maize yields, based on combining crop model simulations with Sentinel-2 imagery, and compares estimates to available field data. Section 5 then applies the maize yields maps to two relevant questions about maize productivity in the region - how spatially variable are maize yields between farmers' fields, and which soil properties appear to be important factors in explaining this variation. Finally, Section 6 discusses the main lessons from this study and implications for future work.

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Study area

Kenya and Tanzania are two of the largest maize producers in sub-Saharan Africa, ranking behind only Nigeria in terms of total maize area, with an official estimate of 2.1 and 4.1 Mha of harvested area and 1.5 and 1.4 t/ha of yield, respectively, in 2017 (FAO, 2018). The typical maize growing systems in both countries are smallholder fields, often intercropped with other staple or cash crops. Inputs level of hybrid seeds and fertilizers are surprisingly low, although intensification is ...

Crop classification

We approached the goal of identifying maize pixels as split into two tasks: one to distinguish cropland pixels (CRL) from non-crop pixels, and one to identify maize pixels (MZL) from other types of crop pixels. Although the final goal was to generate the MZL map, we chose to first generate a CRL map to isolate crop pixels and reduce the overall intra-class variance in the MZL classification process. ...

Scalable crop yield mapper

Following our previous studies (Lobell et al., 2015; Azzari et al., 2017; Burke and Lobell, 2017; Jin et al., 2017b), the scalable crop yield mapper (SCYM) approach was used to estimate maize yields. Briefly, SCYM is a tool that translates satellite-observed vegetation index (VI) and gridded weather product into the maize yields using a regional-specific regression model. Unlike many other empirical methods (Bolton and Friedl, 2013; Johnson, 2014) that requires ground measurements or statistics ...

Applications of yield maps

We present two analytical examples of exploring yield heterogeneity and soil constraints focused on west Kenya. We focus on this region because: (i) satellite imagery was better than other regions in terms of quality and quantity (Fig. S5), which led to greater reliability in yield estimates (Fig. 6d); and (ii) this 40,000 km² is a core production area in Kenya and exhibits substantial yield heterogeneity over the landscape corresponding to a weather gradient (Fig. S10).

...

Discussion

The GEE platform, with its vast stores of satellite data, computational power, and implementation of various machine learning algorithms and flexible user interfaces, was central to this study. For example, GEE provided the possibility to implement a scalable solution to most of Sentinel-2 limitations. Not only we could easily sample and export contaminated pixels from a curated collection of imagery, but we were also able to re-use a light-weight decision tree trained offline to classify ...

Conclusion

This study presented a first-of-its-kind mapping of maize yields at 10m resolution across the entirety of two large maize producing countries, by leveraging the power of newly available satellite sensors, the GEE platform, and advanced classification and yield mapping algorithms. Accomplishing such a mapping in East Africa is extremely challenging because of the heterogeneous smallholder farming landscapes, pervasive and year-round cloud coverage, and the limited and sparse ground ...

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